

MA2040 Jan-Apr 2024

Problem Sheet 1

In the attached file city.pdf you can find the census 2011 data of population, sex ratio and literacy rate of 50 cities of India.

1. Draw the bar graphs of the data sets.
2. Compute the sample mean of population, sex ratio and literacy rate.
3. Compute the sample median of population, sex ratio and literacy rate.
4. Compute the sample mode (or modal values) of population, sex ratio and literacy rate.
5. Compute the sample variance of population, sex ratio and literacy rate.
6. Compute the standard deviation of population, sex ratio and literacy rate.
7. Verify Chebyshev's inequality for $k = 1.5$ of population, sex ratio and literacy rate.
8. Find the sample correlation coefficient between population and literacy rate.
9. Find the sample correlation coefficient between population and sex ratio.
10. Find the sample correlation coefficient between sex ratio and literacy rate.